

Polynomials

A sum of monomials — and the object that the rest of algebra is about.

LESSON 68

1 × 40 MIN

MATEMATIKA 7, §31

S8 2.2

LESSON CLOCK

10'

12'

12'

6'

The definition	10 min
Standard form and degree	12 min
Evaluating	12 min
Homework	6 min

LEARNING OBJECTIVES

- Define a polynomial and name its terms.
- Write a polynomial in standard form and give its degree.
- Name a binomial and a trinomial.
- Evaluate a polynomial for given values.

TEXTBOOK

National	pp. 189–192
Cambridge	Stage 8, pp. 16–20
Workbook	Exercise 2.2

01

Explanation

The definition

A **polynomial** is a sum of monomials. Each monomial is a **term**.

POLYNOMIAL	TERMS	NAME
$3x$	1	monomial
$3x + 5$	2	binomial
$x^2 - 4x + 3$	3	trinomial
$a^3 + a^2 - a + 7$	4	polynomial

Subtraction counts as addition of a negative, so $x^2 - 4x$ is the sum of x^2 and $-4x$.

EVERY MONOMIAL IS A POLYNOMIAL

With one term. The names simply record how many terms there are, and only the first three have special names.

Standard form and degree

A polynomial is in **standard form** when every term is a monomial in standard form, no two terms are similar, and the terms are written in decreasing degree.

AS WRITTEN	STANDARD FORM	DEGREE
$5 + x^2 - 3x$	$x^2 - 3x + 5$	2
$2x + 3x$	$5x$	1
$a^3 + 2a^3 - 4$	$3a^3 - 4$	3
$x^2y + xy^2 + 1$	already standard	3

The **degree of a polynomial** is the greatest of the degrees of its terms. The term of that degree is the **leading term**; a term with no letters is the **constant term**.

COLLECT SIMILAR TERMS BEFORE READING THE DEGREE

In $a^3 + 2a^3 - 4$ the first two terms combine. Reading the degree from an uncollected expression can give the wrong answer when terms cancel.

Evaluating

Substitute and compute, with every negative value in brackets.

POLYNOMIAL	$X = 2$	$X = -1$
$x^2 - 4x + 3$	$4 - 8 + 3 = -1$	$1 + 4 + 3 = 8$
$2x^3 - x$	$16 - 2 = 14$	$-2 + 1 = -1$
$5 - x^2$	$5 - 4 = 1$	$5 - 1 = 4$

A VALUE THAT MAKES THE POLYNOMIAL ZERO IS A ROOT

$x^2 - 4x + 3$ is zero at $x = 1$ and at $x = 3$. Finding roots is what the whole of the equation work later this year is about.

02 Worked examples**EXAMPLE 1** Write $5 + x^2 - 3x$ in standard form and give its degree and constant term.

- 1 Order by decreasing degree.

- 2 $x^2 - 3x + 5$

- 3 Degree 2 — the greatest term degree.

- 4 Constant term 5.

ANSWER $x^2 - 3x + 5$; degree 2, constant 5**EXAMPLE 2** Simplify $a^3 + 2a^3 - 4$ and give its degree.

- 1 Collect the similar terms.

- 2 $3a^3 - 4$

- 3 Two terms — a binomial.

- 4 Degree 3.

ANSWER $3a^3 - 4$, degree 3

EXAMPLE 3 Evaluate $x^2 - 4x + 3$ at $x = 2$ and at $x = -1$.

① $4 - 8 + 3$

② $= -1$

③ $(-1)^2 - 4(-1) + 3 = 1 + 4 + 3$

Brackets round the negative.

④ $= 8$

ANSWER

-1 and 8

03 Interactive model

Ask for the value of $x^2 - 4x + 3$ at $x = 1$ and $x = 3$; both give zero, and the idea of a root arrives before any equation is written.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Polynomial	Ko'phad	Многочлен
Binomial	Ikki had	Двучлен
Trinomial	Uch had	Трёхчлен
Term of a polynomial	Ko'phad hadi	Член многочлена
Degree of a polynomial	Ko'phad darajasi	Степень многочлена
Standard form	Standart ko'rinish	Стандартный вид
Leading term	Bosh had	Старший член
Constant term	Ozod had	Свободный член

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A polynomial is:

a product of monomials

a sum of monomials

a fraction

an equation

$3x + 5$ is a:

monomial

binomial

trinomial

constant

The degree of $x^2 - 3x + 5$:

1

2

3

5

The constant term of $x^2 - 3x + 5$:

1

-3

5

0

$a^3 + 2a^3 - 4$ simplifies to:

$2a^3 - 4$

$3a^3 - 4$

$3a^6 - 4$

$a^3 - 4$

$x^2 - 4x + 3$ at $x = -1$:

0

-1

8

2

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. How many terms in $3x + 5$?

ANSWER 2

2. Its name

ANSWER Binomial

3. How many terms in $x^2 - 4x + 3$?

ANSWER 3

4. Its name

ANSWER Trinomial

5. Degree of $x^2 - 3x + 5$

ANSWER 2

6. Constant term of $x^2 - 3x + 5$

ANSWER 5

7. Degree of 7

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Standard form of $5 + x^2 - 3x$

ANSWER $x^2 - 3x + 5$

2. Simplify $2x + 3x$

ANSWER $5x$

3. Simplify $a^3 + 2a^3 - 4$

ANSWER $3a^3 - 4$

4. Degree of $x^2y + xy^2 + 1$

ANSWER 3

5. $x^2 - 4x + 3$ at $x = 2$

ANSWER -1

6. $x^2 - 4x + 3$ at $x = -1$

ANSWER 8

7. $2x^3 - x$ at $x = 2$

ANSWER 14

Hard · stretch and reason

7 PROBLEMS

1. For which x is $x^2 - 4x + 3 = 0$?

ANSWER $x = 1$ and $x = 3$

2. Simplify $3x^2 - 5x + 2 - x^2 + 5x - 7$

ANSWER $2x^2 - 5$

3. Its degree

ANSWER 2

4. Degree of $(x + 1)(x - 1)$

ANSWER 2

5. $5 - x^2$ at $x = -3$

ANSWER -4

6. A polynomial of degree 0

ANSWER A non-zero constant

7. Simplify $a^2 + a^2 + a^2$

ANSWER $3a^2$

07 Homework

Homework — 5 tasks

Collect similar terms before naming the degree.

1. Write $7 - 2x + x^3$ in standard form and give its degree.
2. Simplify $4y^2 + 3y^2 - 5$ and name the result.
3. Give the constant term of $2x^2 - x + 9$.
4. Evaluate $x^2 + 2x - 5$ at $x = 3$ and at $x = -2$.
5. Find the value of x for which $3x - 6 = 0$.